

Aarya Arban

S11-07

Aim: To implement Packages

Code:

```
package letmecalculate;

public class Calculator {

    public int add (int a, int b)

    {

        return a+b;

    }

    public int multiply (int a, int b)

    {

        return a*b;

    }

    public int subtract (int a , int b)

    {

        return a-b;

    }

    public int divide (int a , int b)

    {

        return a/b;

    }

    public int modulo (int a , int b)

    {

        return a%b;

    }

}

package pattern;

public class PatternMaker {
```

```
public void patternOf(int rows)
```

```
{
```

```
int i,j,k,m;
```

```
for (i=1;i<=rows; i++)
```

```
{
```

```
for (j=1; j<=rows-i; j++)
```

```
{
```

```
System.out.print(" ");
```

```
}
```

```
for (k=1; k<=i; k++)
```

```
{
```

```
System.out.print("*");
```

```
}
```

```
for(m=1;m<i;m++){
```

```
System.out.print("*");
```

```
}
```

```
System.out.println();
```

```
}
```

```
for (i=1;i<=rows; i++)
```

```
{
```

```
for (j=1; j<=i; j++)
```

```
{
```

```
System.out.print(" ");
```

```
}
```

```
for (k=1; k<=rows-i; k++)
```

```
{
```

```
System.out.print("*");
```

```
}
```

```
for(m=1;m<rows-i;m++){
```

```
System.out.print("*");  
}  
System.out.println();  
}  
}  
}  
  
import letmecalculate.*;  
import pattern.*;  
import java.util.*;  
  
public class Demo {  
    public static void main (String[] args)  
    {  
        Calculator random = new Calculator();  
        Scanner s = new Scanner(System.in);  
        int m,n;  
        System.out.println("Enter Number1 : ");  
        m = s.nextInt();  
        System.out.println("Enter Number2 : ");  
        n = s.nextInt();  
        int a = random.add(m,n);  
        int b = random.subtract(m,n);  
        int c = random.multiply(m,n);  
        int d = 0;  
        int e = 0;  
        boolean exc = false;  
        try { d = random.divide(m,n);  
            e = random.modulo(m,n);  
        }  
        catch (ArithmeticException ae)
```

```
{ System.out.println("Denominator cannot be 0");  
exc = true;  
}  
System.out.println("Addition: " + a + "\nSubtraction: " + b + "\nMultiplication: " + c);  
if (!exc) { System.out.println("\nDivision: " + d + "\nModulo: " + e);  
}  
PatternMaker random = new PatternMaker();  
System.out.println("Enter number of rows: ");  
int x = s.nextInt();  
random.patternOf(x);  
}
```